Topic 8: Cell Placement

ECE 9433: Agile System-on-Chip Design

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Cell Placement

- Physically place cells onto your chip
- Optimize their positions and orientations
- Steps
  - Global Placement
  - Detailed Placement

Global Placement

- Place cells in generally the best area using an algorithm
- Minimize total wirelength of the whole design
  - **HPWL**: half-perimeter wirelength
  - Decent proxy for optimizing timing and power
- But also, avoid congestion (cells so close that routing is impossible)

```
[InitialPlace]  Iter: 1 CG residual: 0.00000008 HPWL: 7481451
[InitialPlace]  Iter: 2 CG residual: 0.00000006 HPWL: 6842787
[InitialPlace]  Iter: 3 CG residual: 0.00000009 HPWL: 6812234
[InitialPlace]  Iter: 4 CG residual: 0.00000007 HPWL: 6797416
[InitialPlace]  Iter: 5 CG residual: 0.00000009 HPWL: 6771698
```

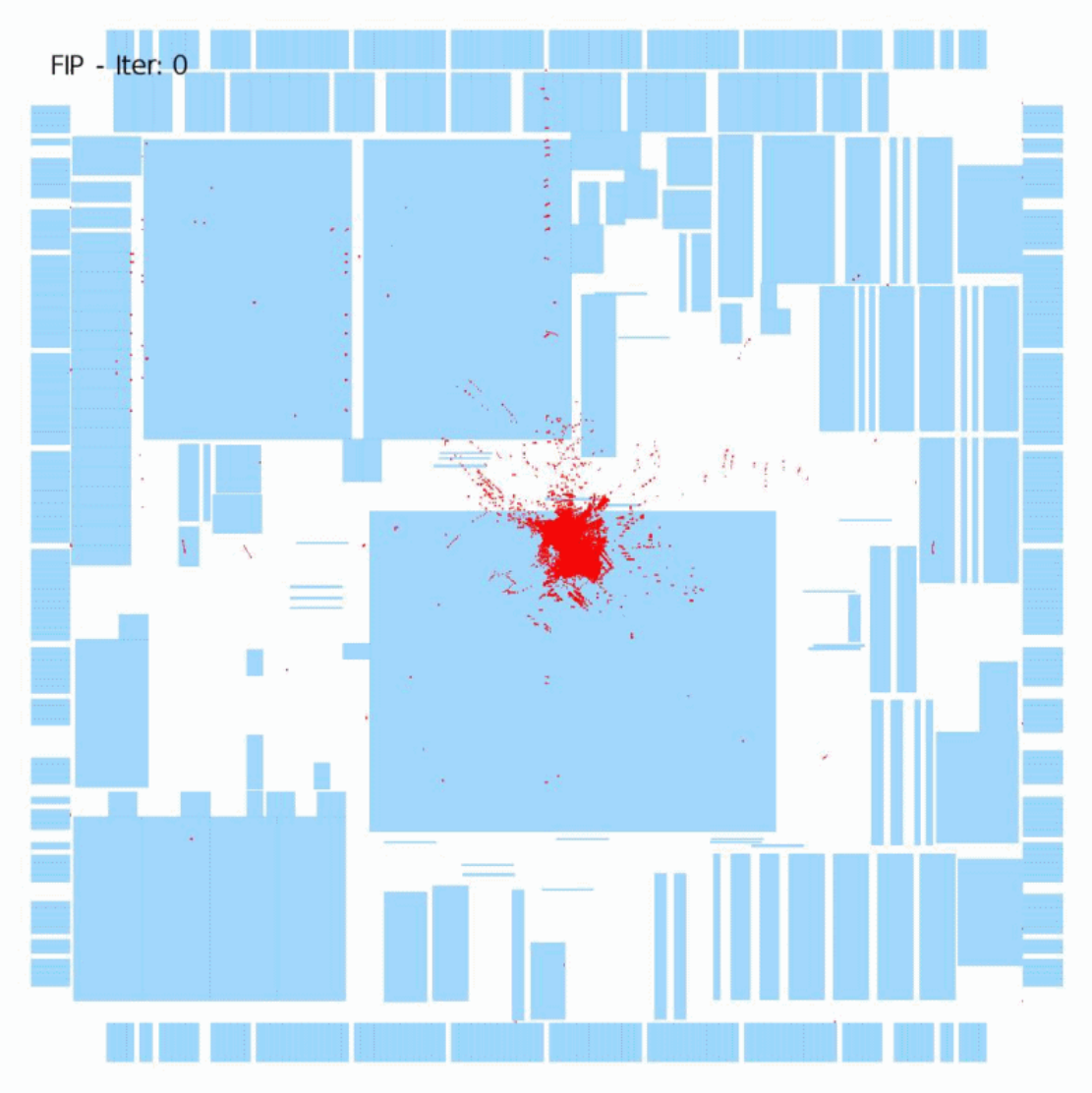
```
[NesterovSolve] Iter: 10 overflow: 0.408459 HPWL: 5518685
[NesterovSolve] Iter: 20 overflow: 0.369914 HPWL: 5301283
[NesterovSolve] Iter: 30 overflow: 0.356915 HPWL: 5274819
[NesterovSolve] Iter: 40 overflow: 0.355273 HPWL: 5251818
[NesterovSolve] Iter: 50 overflow: 0.357892 HPWL: 5262578
# ...
[INFO GPL-0100] worst slack -6.41e-11
[INFO GPL-0103] Weighted 38 nets.
# ...
[NesterovSolve] Iter: 350 overflow: 0.105841 HPWL: 5296714
[NesterovSolve] Finished with Overflow: 0.098895
```

Global Placement → Detailed Placement

Behind the scenes

Global Placement

- Optimize both HPWL and congestion
- Force-directed placement
  - Electrostatic simulation
  - Treat cells as positive charges, empty space as negative charges, connections between cells also as negative charge
- Iterate using gradient descent (yes, similar to machine learning)
- Stop when overflow (congestion) gradient and HPWL gradient go to 0

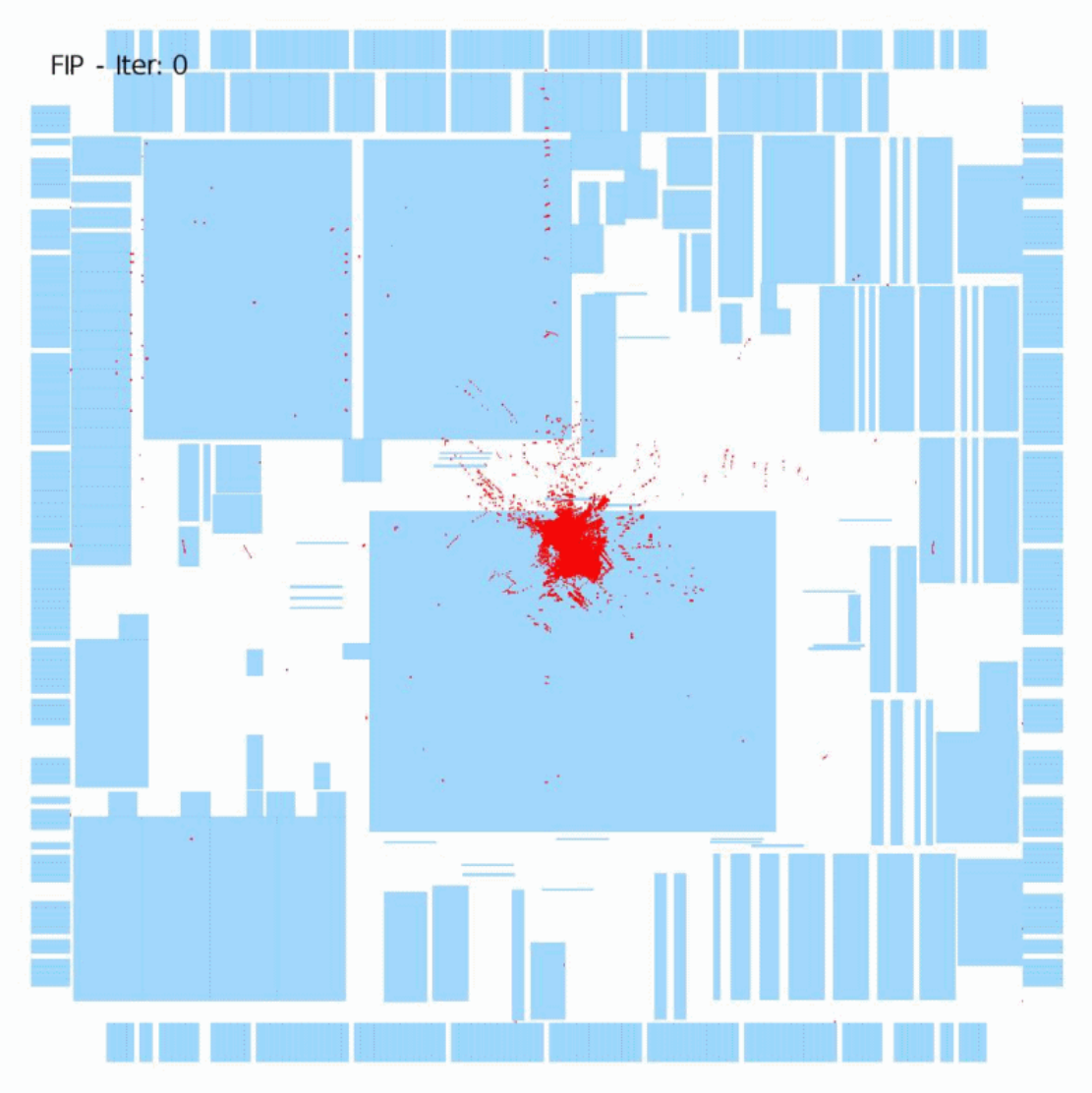


Global Placement Detailed Placement

Behind the scenes

Global Placement

- Timing-driven mode
  - Rebalance charge represented by connections based on actual timing every few iterations
  - Helps remove miscorrelations between HPWL vs. actual circuit timing
- Routability-driven mode
  - Rebalance charge represented by cells based on based on overlapping nets every few iterations
  - Helps remove miscorrelations between placement vs. routing congestion



Global Placement → Detailed Placement

Global Placement

Global Placement Detailed Placement

Detailed Placement

- Legalization: Snap cells to cell sites
- Optimizations
  - Move cells from occupied sites to unoccupied sites if it improves HPWL
  - Swap pairs (or groups) of cells if it improves average HPWL of the group
  - Mirror cells (flip over y-axis) if it reduces HPWL

```
Detailed placement improvement.
Importing netlist into detailed improver.
[INFO DPO-0100] Creating network with 470 cells, 54 terminals, 471 ed
[INFO DPO-0109] Network stats: inst 524, edges 471, pins 1293
[INFO DPO-0110] Number of regions is 1
[INFO DPO-0401] Setting random seed to 1.
# ...
Detailed Improvement Results
----------------------------------------
Original HPWL            2915.0 u
Final HPWL               2703.4 u
Delta HPWL                 -7.3 %
```

```
INFO DPL-0020] Mirrored 20 instances
[INFO DPL-0021] HPWL before          2703.4 u
[INFO DPL-0022] HPWL after           2700.8 u
[INFO DPL-0023] HPWL delta             -0.1 %
```

Global Placement → Detailed Placement

Detailed Placement

Global Placement → **Detailed Placement**

Placement: Common Problems

- Usually, must iterate on area constraints + placement a few times to figure out rough achievable density
- Discontinuities between global and detailed placement
  - Cells stuck in unplaceable regions (placement blockages, macros) will be moved a large amount by legalization
  - Can cause unexpected congestion or unexpectedly bad timing
- Chicken-and-egg problem with timing optimization
  - Timing optimization (next section) can insert cells
  - Inserted cells change congestion
  - Rebalancing congestion changes timing
  - Not easy to get the process to converge for very difficult constraints
- Timing opt. can push to >100% utilization if constraints unreasonable